



Reconstruction and short-term prediction of wind pressure field on cylindrical coal sheds under undisturbed conditions using dynamic mode decomposition

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ABSTRACT

Dynamic Mode Decomposition (DMD) accurately captures growth rates and frequency characteristics of each mode, establishing a reduced-order model for the evolution of the flow field to reconstruct or predict the flow dynamics process. This study applies DMD to analyze the wind pressure distribution problem of cylindrical coal sheds, investigating its accuracy in analyzing pressure fields. First, wind tunnel tests were conducted on cylindrical coal sheds to examine the mean wind pressure coefficient distribution under typical wind directions (defined as θ). Subsequently, the wind pressure field of the coal shed was decomposed using the DMD at $\theta = 0^\circ$. Reconstruction with the first 15 DMD modes can better characterize the wind pressure distribution. Therefore, reconstruction with the first 15 DMD modes was chosen to reconstruct the pressure fields conducted at $\theta = 30^\circ$, 60° , and 90° . The DMD decomposition performance is optimal at $\theta = 0^\circ$ for the cylindrical surface. For pressure field reconstruction in the same wind direction, as the number of modes increases, the reconstruction relative error of the pressure field decreases, thereby enhancing reconstruction accuracy and the ability to capture details. Finally, short-term predictions of the coal shed's wind pressure distribution were conducted, revealing better predictive performance near the incoming flow position. DMD can provide a basis for studying the wind pressure distribution on large-span structures such as coal sheds.

1. Introduction

Due to the advancement of economy and technology, the emergence of new building materials continuously propels the progress of construction, where architectural structures are gradually transitioning towards lightweight and large-span directions. Large-span spatial structures have experienced rapid development, yet the application of lightweight materials has made these structures increasingly sensitive to wind excitation. The wind effects on large-span spatial structures are immensely intricate, thus wind load stands as a principal control when designing such structures (Yasui et al., 1999). Coal sheds exemplify typical large-span spatial structures, characterized by their light weight, high flexibility, and low natural frequency (Su et al., 2018). Moreover, located in the turbulent near-ground region of the atmospheric boundary layer, coal sheds are subject to exceedingly complex aerodynamics

and wind-induced damages, which could result in substantial economic losses and environmental pollution. The study of the wind pressure distribution and wind loads on the surface of coal sheds is of great importance.

Currently, wind tunnel tests are predominantly used to study the wind load distribution on cylindrical grid shells and other large-span curved grid shell structures (Qiu et al., 2014; Su et al., 2018; Su et al., 2020). Luo et al. (2021) investigated the characteristics of wind pressure on typical large-scale spatial structures through wind tunnel tests and field measurements. Li et al. (2006) measured the wind pressure distribution of cylindrical shell models with varying shape ratios in a wind tunnel. Through equivalent static and dynamic analyses, they examined the wind load distribution and its influence on the widely used single-layer mesh cylindrical shells. In turbulent boundary layer flow, Chen et al. (2021) characterized the wind pressure distribution over a

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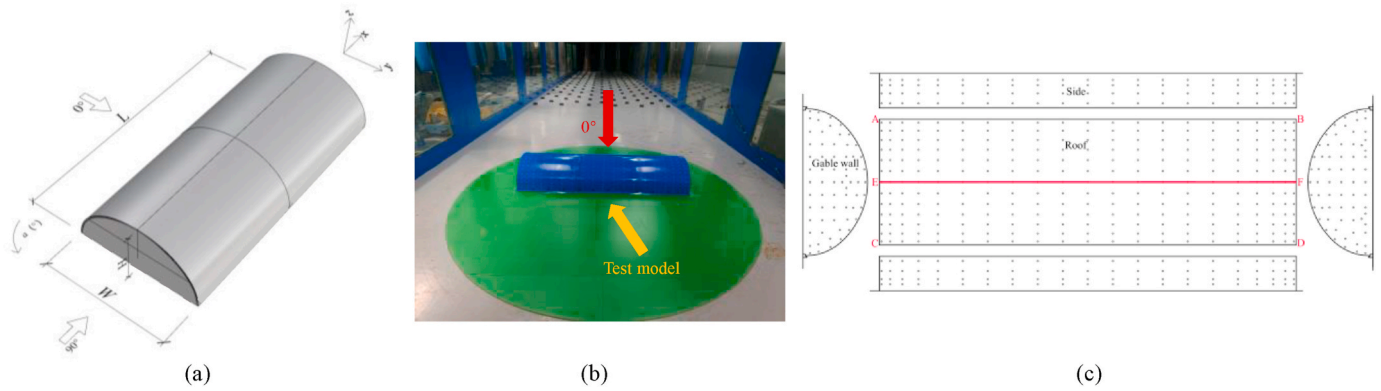


Fig. 1. Relevant parameters of the experimental model: (a) model parameters and wind directions; (b) experimental model in the wind tunnel; (c) arrangement of measuring points.

cantilevered roof by testing a 1/100 scale model of the roof over a grandstand.

The flow field around cylindrical coal sheds exhibits complex phenomena such as airflow separation, reattachment, and vortex shedding, making the distribution of wind pressure around cylinders highly intricate. Further understanding of the intricate flow characteristics on the surface is necessary. The data gathered from wind tunnel tests contain a large amount of information in spatial and temporal dimensions. Hence, obtaining flow characteristics under low-dimensional space from the high-dimensional nonlinear flow field and establishing Reduced-Order Models (ROMs) to enhance efficiency and performance has become a prominent research focus. Proper Orthogonal Decomposition (POD) and Dynamic Mode Decomposition (DMD) are two typical methods (Kou and Zhang, 2017; Li et al., 2022).

POD offers an effective approach for describing random fields by decomposing the wind pressure field into a time-only principal coordinate and space-only characteristic vectors (Holmes et al., 1997). To describe the wind pressure field more accurately, only a small number of terms are required in the expansion. POD has been widely introduced by many researchers in the wind engineering field and extensively applied to the study of fluctuating wind loads on low-rise buildings (Holmes et al., 1997; Bienkiewicz et al., 1995), high-rise buildings (Kikuchi et al., 1997; Kim et al., 2018), and the pulsating wind pressure affecting the dynamic response of bridge structures (Ricciardelli and Grenet, 2002). However, POD also has certain limitations. It assesses modal contributions based on energy content rather than dynamic influence, failing to account for low-energy modes that may have a significant dynamic impact on the dataset (Kou and Zhang, 2017). Additionally, to maintain the orthogonality of the spatial modes, the resulting POD modes can contain multiple frequencies, which may not be suitable for interpreting physical phenomena.

DMD proposed by Schmid (2010), is a data-driven ROM that accounts for both the temporal and spatial correlations of the flow field sequence. It extracts the dynamic structures ingrained in the flow field from high-dimensional flow data, subsequently establishing a foundational low-order dynamical model of the flow field. Based on Koopman operator theory, DMD approximates the flow evolution process as a linear dynamical system. By analyzing snapshots of the entire flow field process, DMD identifies eigenvalues and eigenvectors corresponding to each mode, yielding low-order modes of flow field information (Zhou et al., 2021; Schmid, 2022). By combining these eigenvalues and eigenvectors, DMD can accurately reconstruct the temporal evolution of the flow system, revealing its spatiotemporal characteristics. In contrast to POD, which focuses on energy content, DMD provides individual frequencies and growth rates for each mode, enabling a detailed analysis of specific flow behaviors and their variations (Bistrrian and Navon, 2017; Fu et al., 2023).

DMD has been widely applied across various disciplines, with fluid

mechanics being particularly prominent (Zhang et al., 2014). Zhi et al. (2023) employed large eddy simulation (LES) to obtain the transient vortex wake structure of a propeller. Through DMD analysis and reconstruction, they observed that the coherent structures determined by DMD exhibited low correlation with the hub vortices, but were mainly related to organized convection in the tip vortices. Qiu et al. (2020) numerically investigated the transient cavitation flow around a Clark-Y airfoil using DMD. They found that the frequencies of the first four modes identified by DMD were consistent with those obtained from the Fast Fourier Transform (FFT), and subsequently reconstructed the cavitation flow field using these four modes, thereby capturing the primary flow features. Bai et al. (2020) combined numerical simulations with DMD to analyze the stability and flow characteristics of parallel double cylinders, revealing the fundamental physical principles of the deflection oscillatory turbulent mode at low Reynolds numbers and moderate gap ratios.

Despite its significant promise, DMD remains a relatively new concept within the field of wind engineering (Li et al., 2023). Zhang et al. (2014) conducted a comprehensive comparison of the effectiveness of POD and DMD in identifying coherent structures within the wake flows of cylindrical objects of different diameters under varying Reynolds number conditions. The comparative results revealed that the DMD is capable of distinctly separating each structure in both spatial and spectral domains, whereas the corresponding POD modes associated with the target structures may suffer contamination from irrelevant structures. To gain insights into the alleviation of vortex shedding from bridge models by passive control vortex generators, Zhang et al. (2019) employed the DMD to extract prominent flow field characteristics and identify wake modes and their corresponding frequencies. Their conclusion indicated that vortices can redistribute energy spectra, thereby altering the frequency of vortex shedding. At a Reynolds number of 22,000 Li et al. (2020) applied DMD to a square prism, studying its accuracy and practical value in analyzing pressure fields and its performance in data reconstruction.

Flow field reconstruction serves as an excellent method for evaluating the accuracy of ROMs. POD requires computational steps such as matrix operations and singular value decomposition, making it comparatively intricate for flow field reconstruction. In contrast, DMD utilizes matrix decomposition techniques, which are computationally more convenient and efficient during the reconstruction process. Meanwhile, the accuracy of the DMD algorithm can also be verified by reconstructing the flow field with high fidelity (Li et al., 2020). In this paper, the flow field reconstruction and the absolute error between the original and reconstructed flow fields are chosen as the evaluation scheme of the DMD algorithm.

It should be noted that, in this study, the wind acts directly on the surface of the measured coal shed, and the influence of surrounding structures on the mean wind pressure coefficient of the coal shed is

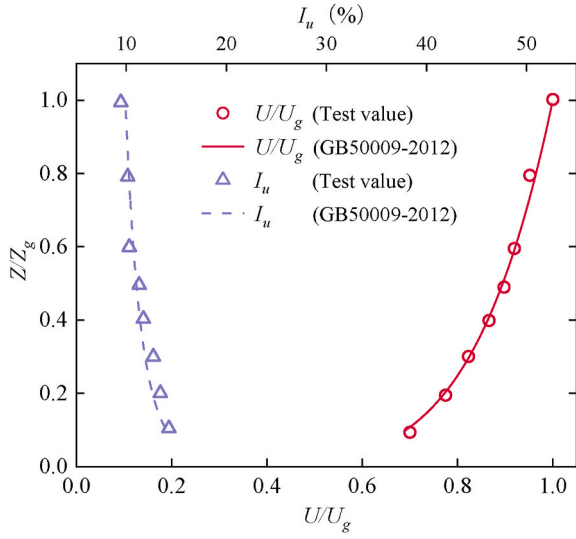


Fig. 2. Mean wind speed and turbulence intensity profiles of the wind tunnel test.

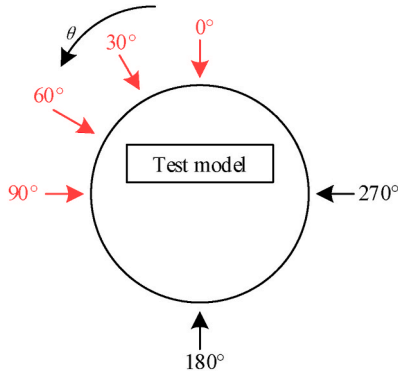


Fig. 3. Schematic diagram of wind directions.

relatively small. Therefore, we can approximately treat the coal shed as being in an undisturbed environment. A single enclosed cylindrical coal shed was chosen as the rigid model for synchronous pressure measurement tests conducted in a wind tunnel within the atmospheric boundary layer. The rest of this paper is organized as follows. Section 2 provides an introduction to the wind tunnel test of the coal shed and presents the fundamental equations of the DMD algorithm. In Section 3, an analysis of the experimental results and the reconstruction effect of DMD is conducted, along with an investigation into its short-term predictive capabilities. Finally, Section 4 provides a summary of the content presented in this paper.

2. Methodology

2.1. Wind tunnel test

The wind tunnel test was conducted in the low-speed test section of the STU-1 wind tunnel laboratory at the Wind Engineering Research Center of Shijiazhuang Tiedao University. The dimensions of this test section are 4.4 m in width, 3.0 m in height, and 24.0 m in length, with a maximum wind speed of 30.0 m/s. This study is based on a real coal shed, with dimensions of approximately 280 m in length, 100 m in width, and a height of about 45 m. A scaled model of the coal shed was tested in the wind tunnel, with a scale ratio of 1:150. The model, constructed from ABS plastic, has a longitudinal length (L) of 1.9 m, a span (W) of 0.7 m, and a height (H) of 0.3 m. The length-to-span ratio is 2.71,

while the height-to-span ratio is 0.43. A total of 572 locations were equipped with double-sided pressure measurement points in the model. Specifically, 273 points were arranged on the roof of the coal shed, 168 points on the sides, and 131 points on the gable walls. The overall number of measurement points on the model reached 1144. The experimental model installed in the wind tunnel and the locations of the pressure measurement points are illustrated in Fig. 1.

The wind tunnel test of the model was carried out under non-interference conditions, with the surrounding terrain classified as exposure B category (type B in GB 50009-2012, (National Standards Committee, 2012)). To simulate the wind profile of terrain type B, spires and roughness elements were utilized. Fig. 2 shows the mean wind speed and turbulence intensity profile simulated in the experiment are in good agreement with the theoretical values specified in the code. According to the design code, the formulas for calculating the mean wind speed and turbulence intensity are given by equations (1) and (2), respectively.

$$\frac{U(z)}{U_g} = \left(\frac{z}{Z_g} \right)^\alpha \quad (1)$$

$$I_u(z) = I_g \left(\frac{z}{Z_g} \right)^{-\alpha} \quad (2)$$

where z represents the height of the measurement; Z_g represents the reference height of 10 m; $U(z)$ is the mean wind speed at height z ; U_g is the mean wind speed at the reference height of 10 m (10 m above the ground); $I_u(z)$ is the turbulence intensity at height z ; I_g is the turbulence intensity at the reference height of 10 m, and α is the wind profile index related to surface roughness, in this article $\alpha = 0.16$.

The experiment was conducted under undisturbed B-type terrain conditions with a free-flow wind speed of 16 m/s. The pressure scanning valve had a sampling frequency of 330 Hz and a sampling time of 30 s. The experiment defined the incident inflow that is perpendicular to the longitudinal axis of the cylindrical coal shed as a wind direction of 0° , incrementally rotating counterclockwise at intervals of 15° . Wind direction ranged from 0° to 360° , as defined in Fig. 3, with the wind direction represented by θ .

2.2. Data preprocessing

The wind pressure data gathered from the experiment represent the recorded pressure values at multiple measurement points distributed across the surface of the model. To simplify the analysis, the pressure values at each measurement point on the structural surface are converted into dimensionless pressure coefficients, commonly referred to as wind pressure coefficients. The formula for the wind pressure coefficient is as follows:

$$C_{pi}(t) = \frac{P_i(t) - P_s}{P_t - P_s} \quad (3)$$

where $C_{pi}(t)$ represents the wind pressure coefficient at the location of the i th pressure measurement point on the model, $P_i(t)$ denotes the instantaneous wind pressure value obtained at that location, P_s represents the mean static pressure at the reference point, and P_t signifies the mean total pressure at the reference point. The height of the reference point is taken as a height of 10 m.

2.3. Dynamic mode decomposition (DMD)

DMD is a data-driven approach used for analyzing and simulating dynamic systems, which finds extensive application in low-dimensional reduction analysis of high-dimensional complex flow fields (Bistrian and Navon, 2017; Alekseev et al., 2016). The major steps involved in the DMD algorithm are explained as follows (Schmid, 2010; Li et al., 2020).

The pressure field data acquired from the experiment are processed

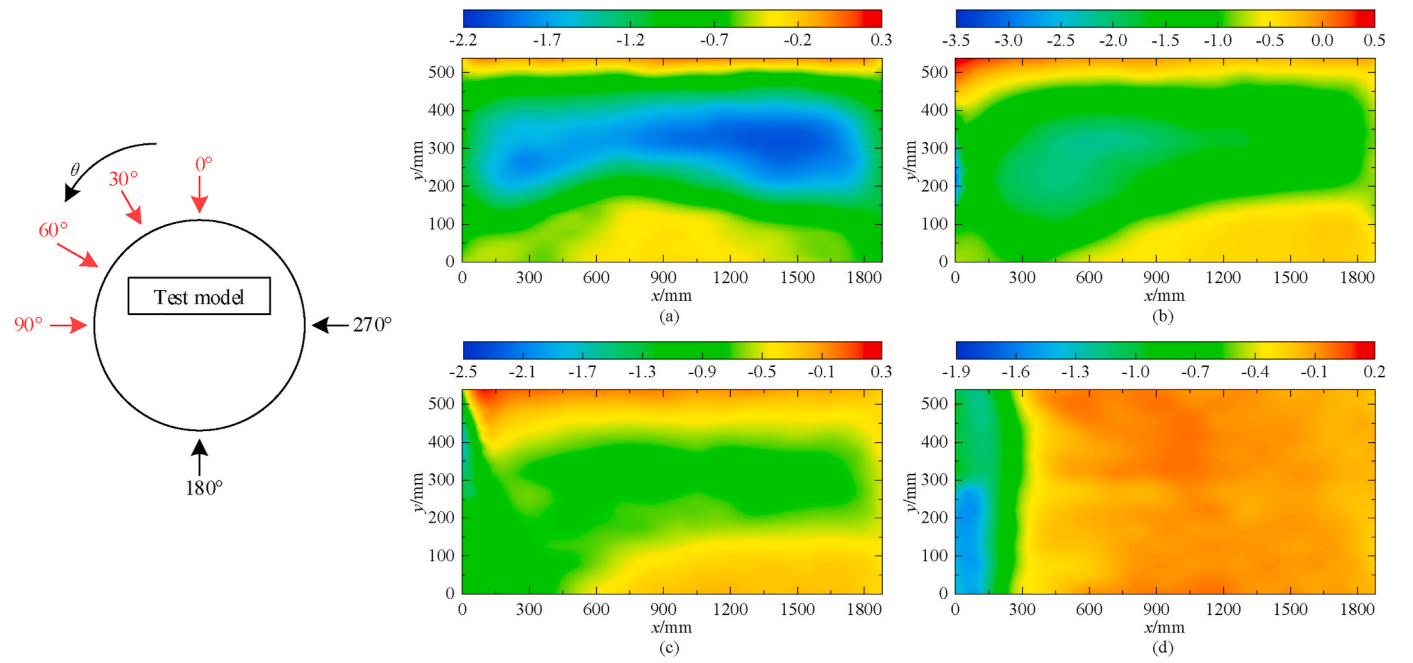


Fig. 4. Contour plots of mean wind pressure coefficients: (a) 0°, (b) 30°, (c) 60°, (d) 90°.

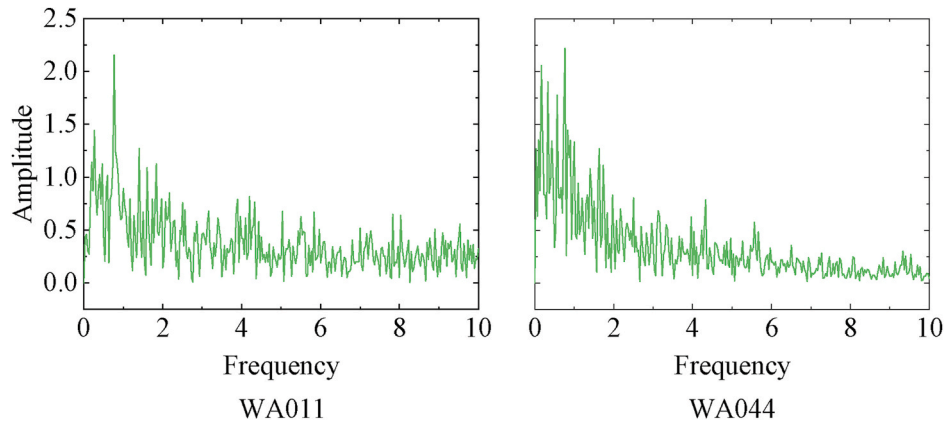


Fig. 5. Wind pressure frequency spectra at $\theta = 0^\circ$.

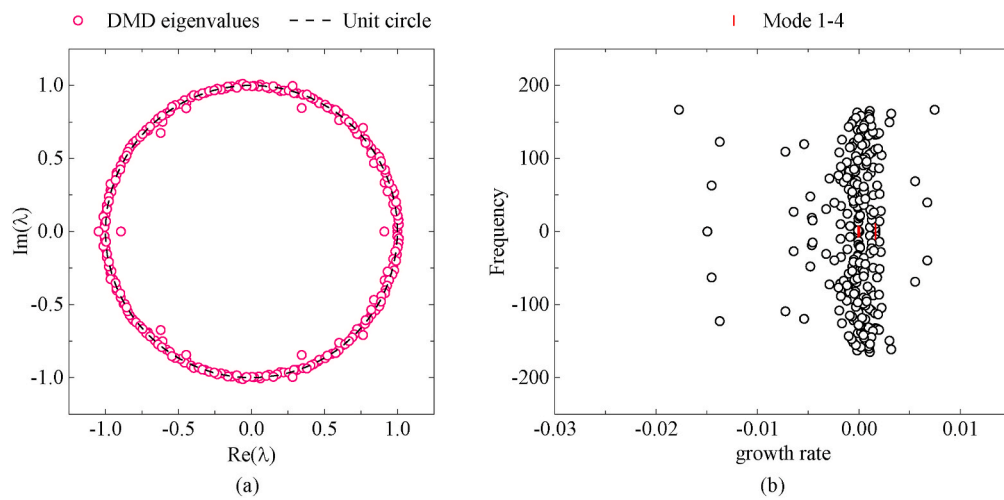


Fig. 6. (a) The mode eigenvalues at $\theta = 0^\circ$, (b) the growth rate distribution of DMD.

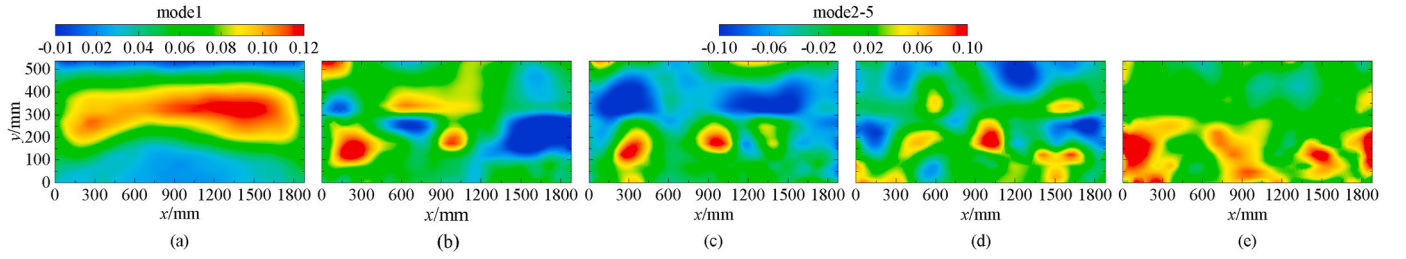


Fig. 7. The first 5 DMD modes of the pressure field.

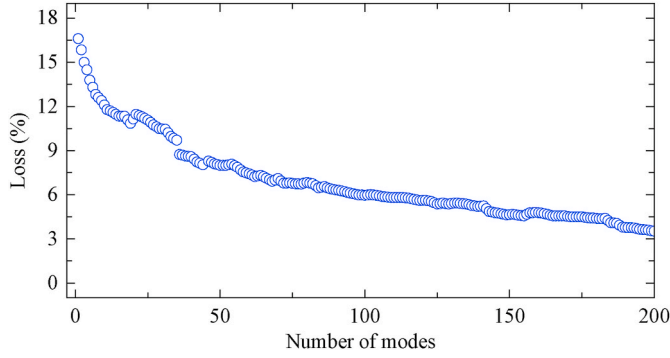


Fig. 8. The correlation between the loss and the number of modes.

to obtain a series of snapshots at N discrete time instances, arranged in the sequence from 1 to N . We will get a snapshot matrix $\mathbf{CP}_1^N = [Cp_1 \ Cp_2 \ \dots \ Cp_N]$. Δt denotes the time interval between any two adjacent snapshots. Subsequently, two snapshot matrices, $\mathbf{CP}_1$ and $\mathbf{CP}_2$, are constructed as follows:

$$\mathbf{CP}_1 = [Cp_1 \ Cp_2 \ \dots \ Cp_{N-1}] \quad (4)$$

$$\mathbf{CP}_2 = [Cp_2 \ Cp_3 \ \dots \ Cp_N] \quad (5)$$

where Cp_i represents the data snapshot of the i^{th} pressure field.

Assuming a linear approximation exists between the snapshot matrices $\mathbf{CP}_1$ and $\mathbf{CP}_2$, it can be represented as:

$$\mathbf{CP}_2 = \mathbf{A} \mathbf{CP}_1 \quad (6)$$

where $\mathbf{A}$ represents the optimal fitting linear approximation between

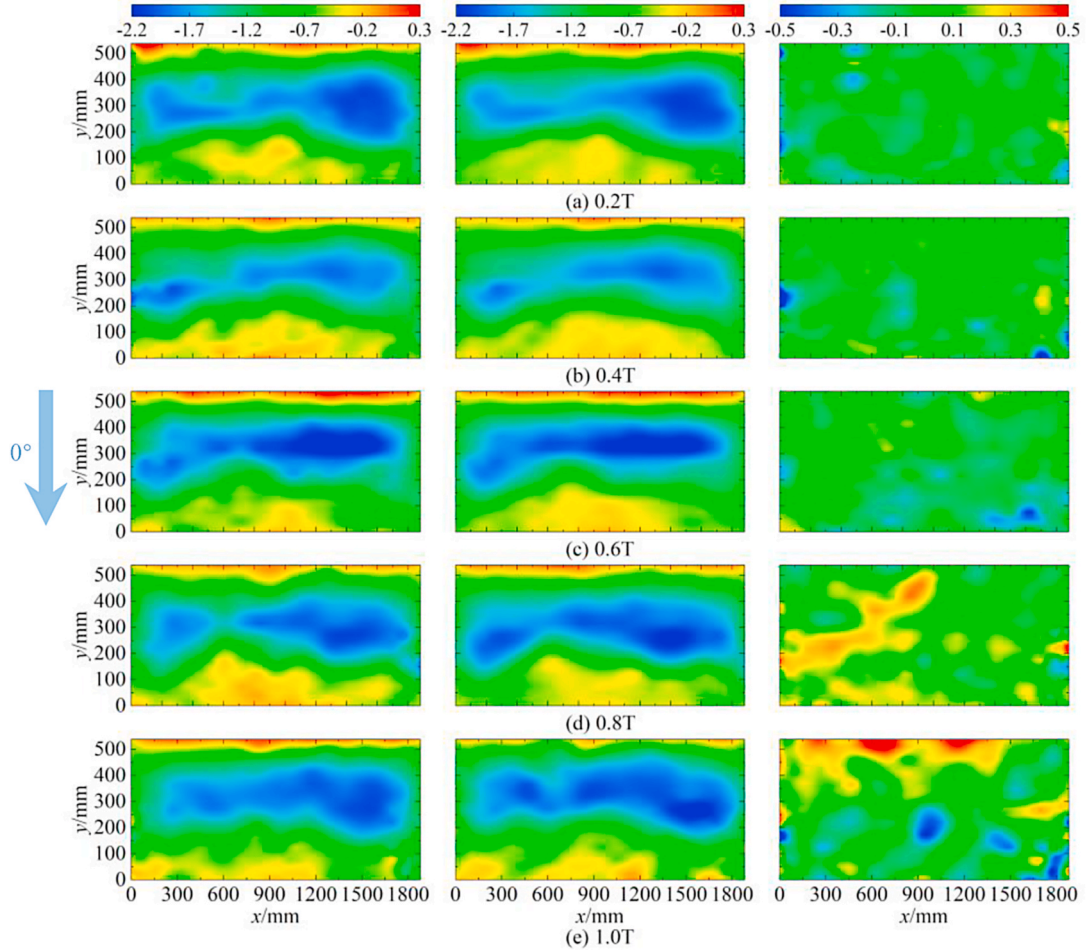


Fig. 9. The original and reconstructed pressure field corresponding to the first 5 modes at five instants (Left: original pressure field; middle: reconstructed pressure field; right: absolute error of the reconstruction).

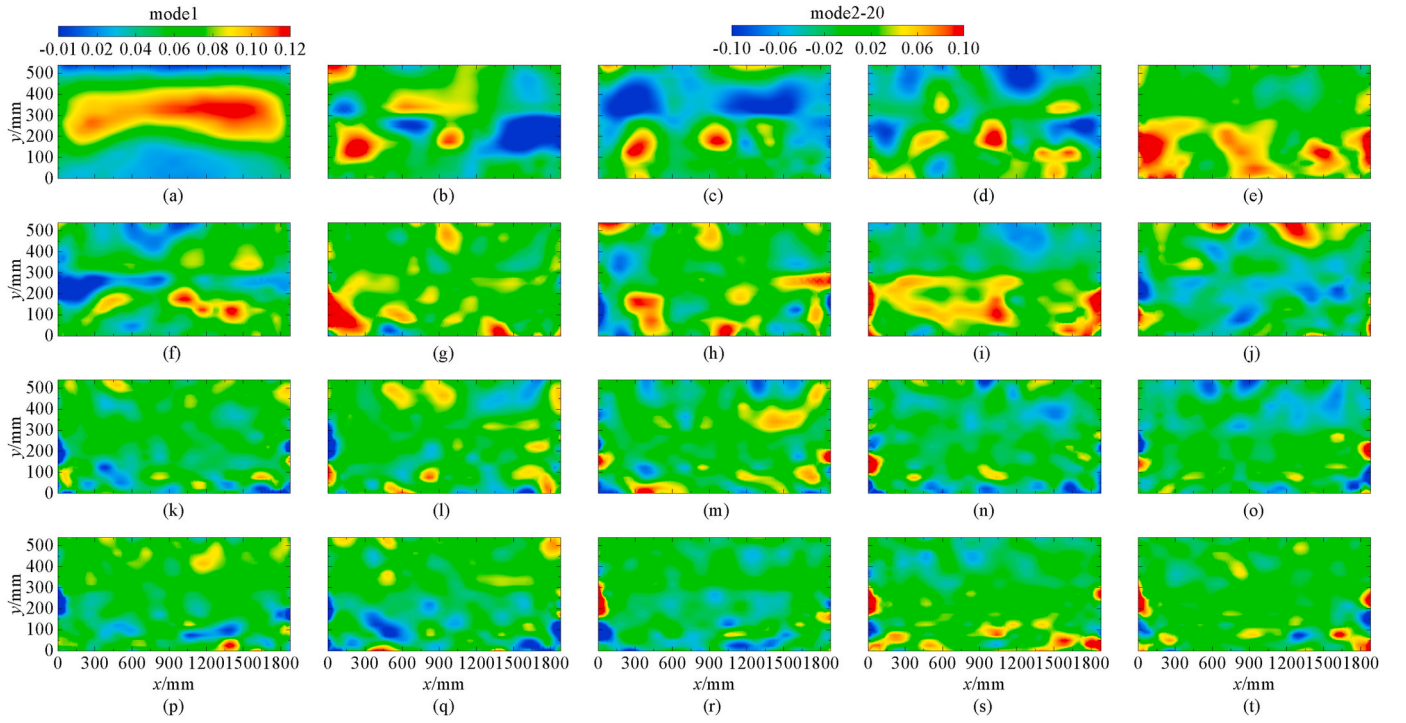


Fig. 10. The first 20 DMD modes of the pressure field.

$\mathbf{CP}_1$ and $\mathbf{CP}_2$. In practice, the computation of high-dimensional matrix $\mathbf{A}$ can be challenging for large datasets, necessitating the derivation of a similar matrix expression (Tu, 2013). Firstly, $\mathbf{CP}_1$ is subjected to Singular Value Decomposition (SVD) (Halko et al., 2011), projecting it onto a lower-rank subspace.

$$\mathbf{CP}_1 = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (7)$$

where superscript T denotes the conjugate transpose, $\mathbf{U} \in \mathbb{C}^{N \times r}$, $\mathbf{U}^T \mathbf{U} = \mathbf{I}$, $\mathbf{V} \in \mathbb{C}^{M \times r}$, $\mathbf{V}^T \mathbf{V} = \mathbf{I}$, $\mathbf{I}$ denotes the identity matrix, $\mathbf{\Sigma}$ represents a diagonal matrix $\mathbf{\Sigma} \in \mathbb{C}^{r \times r}$, with its diagonal elements arranged from top to bottom in ascending order of singular values, and r indicates the truncated rank for simplified SVD.

Then, a lower-dimensional matrix $\tilde{\mathbf{A}} \in \mathbb{C}^{r \times r}$ similar to $\mathbf{A}$ is constructed to replace the high-dimensional matrix $\mathbf{A}$ (Kutz et al., 2016). Estimate the eigenvalues of $\mathbf{A}$ using the eigenvalues of $\tilde{\mathbf{A}}$.

$$\mathbf{A} = \mathbf{U}\tilde{\mathbf{A}}\mathbf{U}^T \quad (8)$$

The computation of the matrix $\tilde{\mathbf{A}}$ can be conceptualized as a minimization problem between $\mathbf{CP}_1$ and $\mathbf{CP}_2$.

$$\underset{\tilde{\mathbf{A}}}{\text{minimize}} \|\mathbf{CP}_2 - \mathbf{A}\mathbf{CP}_1\|_F^2 \quad (9)$$

At the same time, it can be represented as:

$$\underset{\tilde{\mathbf{A}}}{\text{minimize}} \|\mathbf{CP}_2 - \mathbf{U}\tilde{\mathbf{A}}\mathbf{V}^T\|_F^2 \quad (10)$$

where $\|\cdot\|_F$ denotes the Frobenius function, and $\mathbf{A}$ can be approximated as $\tilde{\mathbf{A}}$.

$$\mathbf{A} \approx \tilde{\mathbf{A}} = \mathbf{U}^T \mathbf{CP}_2 \mathbf{V} \mathbf{\Sigma}^{-1} \quad (11)$$

Next, the eigenvalue decomposition of the matrix $\tilde{\mathbf{A}}$ is calculated using the following equation:

$$\tilde{\mathbf{A}}\mathbf{W} = \mathbf{W}\mathbf{\Lambda} \quad (12)$$

where the columns of matrix $\mathbf{W}$ represent the eigenvectors w_j , and

matrix $\mathbf{\Lambda}$ is a diagonal matrix with the eigenvalues λ_j on its diagonal.

If the matrix $\tilde{\mathbf{A}}$ encompasses the dominant eigenvalues of $\mathbf{A}$, then the j^{th} eigenvector of $\mathbf{A}$ (or the j^{th} DMD mode) is given by:

$$\Phi_j = \mathbf{U}w_j \quad (13)$$

where Φ_j corresponds to the j^{th} DMD mode, and w_j corresponds to the eigenvector corresponding to the j^{th} eigenvalue of the matrix.

The frequency ω_j and growth rate (decay rate) g_j of the j^{th} DMD mode can be computed as follows:

$$\omega_j = \text{Im}\{\log(\lambda_j)\} / \Delta t \quad (14)$$

$$g_j = \text{Re}\{\log(\lambda_j)\} / \Delta t \quad (15)$$

By employing the aforementioned DMD, the dynamic modes of the pressure field can be extracted. Subsequently, the reconstructed pressure field can be generated using the obtained DMD modes, which are sorted based on the mode amplitudes. The instantaneous pressure field at any instant in the time series is given by the following formula:

$$Cp(i) \approx \sum_{k=1}^r \varphi_k(\mu_k)^{i-1} \alpha_k \quad (16)$$

where φ_k represents the k^{th} DMD mode, μ_k denotes the k^{th} eigenvalue, and $\alpha_k = \Phi_k^\dagger v_1$ signifies the amplitude of the k^{th} mode, representing its contribution to the initial snapshot Cp_1 , $\dagger$ indicates the pseudo-inverse.

3. Results and discussions

3.1. Characteristics of the pressure field

The data used in this study were derived from wind tunnel tests. The original data obtained from the experiment represent the wind pressure values at various pressure measurement points on the surface of the model. To facilitate expression, the wind pressure data have been transformed into non-dimensional wind pressure coefficients denoted by Cp . In this study, when defining the wind pressure acting on the surface

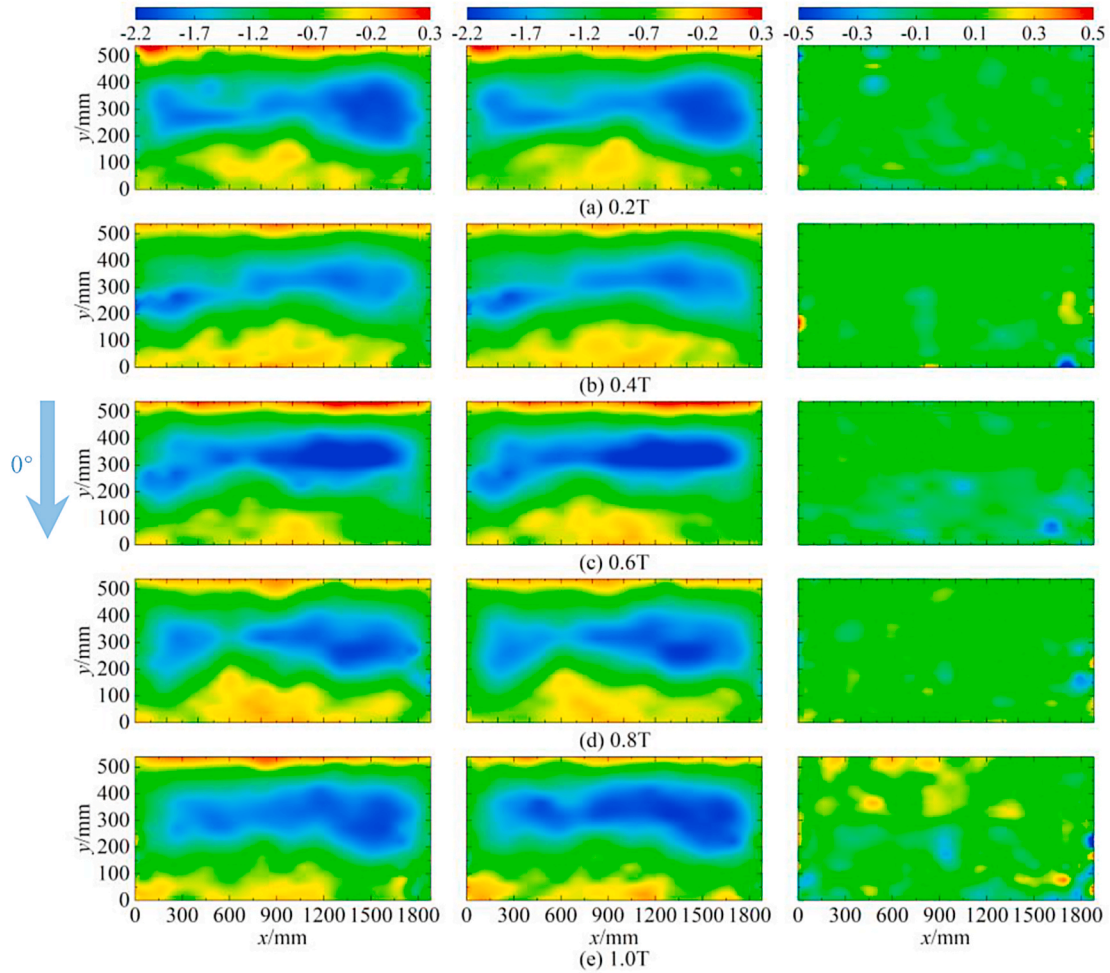


Fig. 11. The original and reconstructed pressure fields corresponding to the first 15 modes at five instants (Left: original pressure field; middle: reconstructed pressure field; right: absolute error of the reconstruction).

of the structure, positive pressure indicates forces directed toward the interior of the structure, while negative pressure indicates forces directed away from it. The terms “positive” and “negative” solely represent direction and do not signify numerical magnitude. Positive pressure represents the action of wind pressure force, whereas negative pressure represents the action of wind suction force. The roof area ABCD of the model is selected as the research object, with a total of 273 pressure measurement points distributed as shown in Fig. 1(c). A total of 250 sampling instants were selected for the pressure measurement points on the surface of the model to calculate the mean wind pressure coefficients. The test includes various wind directions, and the working conditions are complex, which complicates the comparison of the data. At $\theta = 0^\circ$, the front edge AB of the surface contacts the incoming flow first. When $\theta = 30^\circ$, the edges AB and AC of the surface come into contact with the incoming flow. At $\theta = 90^\circ$, the edge AC of the side of the structure collides with the incoming flow. Therefore, the analysis focuses on the characteristic distribution of wind pressure coefficients on the surface at representative wind directions of 0° , 30° , 60° , and 90° . Fig. 4 presents contour plots of mean wind pressure coefficients for a closed cylindrical coal shed at various wind directions.

At $\theta = 0^\circ$, Fig. 4(a) demonstrates that the wind pressure coefficients exhibit good symmetry in all regions except for the top area. In the vicinity of the edge AB of the cylindrical coal shed, there is positive pressure. As the height increases, the wind pressure coefficients decrease and gradually become negative. A negative pressure zone is formed in the rooftop area, where the mean wind pressure coefficient reaches the maximum negative value of -2.03 . On the leeward side EFCD of the coal

shed, the absolute values of negative pressure gradually diminish. As the airflow approaches the surface of the coal shed, separation occurs, leading to the formation of discrete vortices through some not yet fully understood processes. These vortices detach into the wake on the lee side of the coal shed, generating significant suction forces near the separation point.

Fig. 4(b) illustrates the contour plot of mean wind pressure coefficients at $\theta = 30^\circ$. When the airflow encounters the edges AB and AE of the cylindrical surface, separation phenomena occur. At the edge AB, in the red area, the airflow flows along the edge of the coal shed, resulting in a positive mean wind pressure coefficient, which manifests as a wind pressure force. At the green region near edge AC, flow separation takes place due to the adverse pressure gradient, causing suction on the surface of the coal shed. In other regions, particularly the top of the coal shed, negative pressure zones are still present, and the maximum absolute value of the mean wind pressure coefficient of negative pressure reaches 3.33.

At $\theta = 60^\circ$, the distribution of wind pressure resembles that at $\theta = 30^\circ$. The positive wind pressure coefficient primarily occurs in the AE area of the cylinder edge, with a concentration towards the red area of the AB edge, reaching a maximum positive mean value of approximately 0.19.

At $\theta = 90^\circ$, due to the collision between the incoming flow and the structural edge AC, the mean wind pressure coefficient in the AC region of the cylindrical end primarily exhibits negative values, indicating a wind suction effect. In the middle and back parts of the cylindrical coal shed, with the occurrence of flow separation and reattachment, the

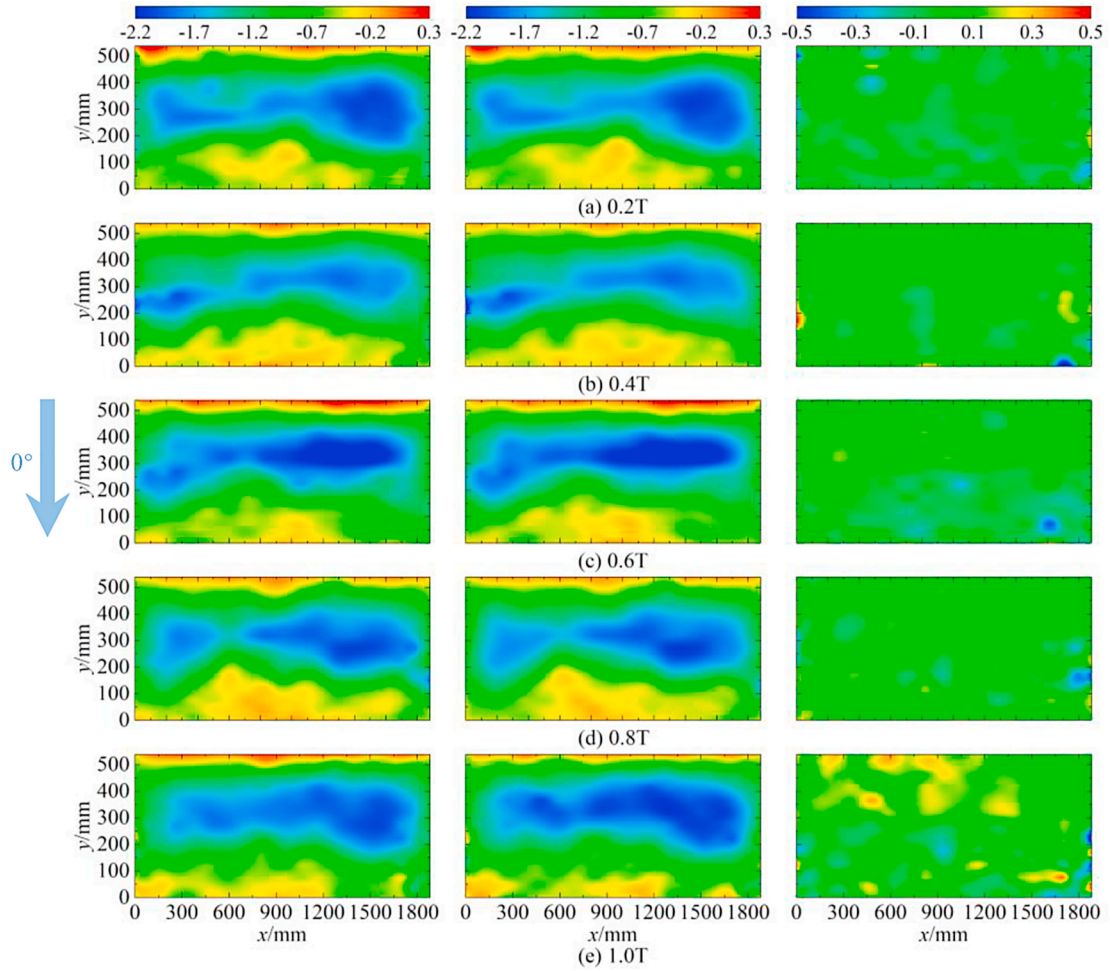


Fig. 12. The original and reconstructed pressure fields corresponding to the first 20 modes at five instants (Left: original pressure field; middle: reconstructed pressure field; right: absolute error of the reconstruction).

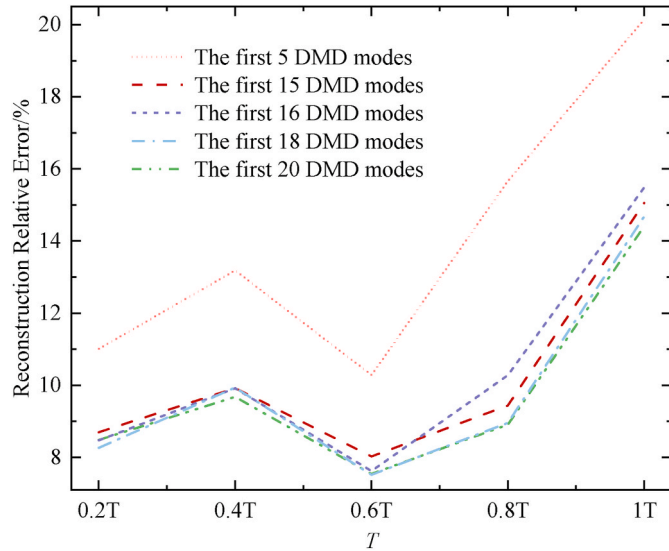


Fig. 13. The reconstruction relative errors of modal reconstructions at various orders.

mean wind pressure coefficients on the surface of the coal shed structure are relatively close and are distributed more uniformly.

The resonant frequency of the coal shed structure is low, while the

roof structure exhibits a dense distribution of frequencies. Moreover, a significant portion of the roof is situated within regions of airflow separation and reattachment. The wind pressure field on the surface of the cylinder exhibits pronounced spatial and spatiotemporal variability, with considerable uncertainty in its spatial distribution (Su et al., 2018). Obtaining the low-dimensional flow characteristics from a high-dimensional nonlinear flow field is necessary to further understand the wind pressure distribution on the surface of the cylinder. Therefore, DMD is employed for further analysis of the wind pressure field on the cylindrical surface of the coal shed.

3.2. DMD analysis and reconstruction at wind direction of 0°

Fourier transform analysis was performed on the data of measurement points at $\theta = 0^\circ$. Fig. 5 illustrates the wind pressure frequency spectra for two typical measurement points (WA011 and WA044). In these spectra, the low-frequency range exhibits higher energy, with multiple frequency components densely distributed. These frequencies have a more direct impact on the dynamic response of the structure. Consequently, the frequency range of 0–10 Hz was selected for display. At $\theta = 0^\circ$, the dominant frequency is 0.78 Hz, with a corresponding cycle of 1.28 s. To improve the reconstruction and prediction accuracy of the DMD, data corresponding to five complete cycles were selected for analysis, resulting in a total time span of 6.4 s. Each cycle contains 50 data points, so the pressure field data of $T = 250$ time points are selected in total. During data processing, to ensure that the time interval of the pressure field data matches the time interval of the snapshot, the

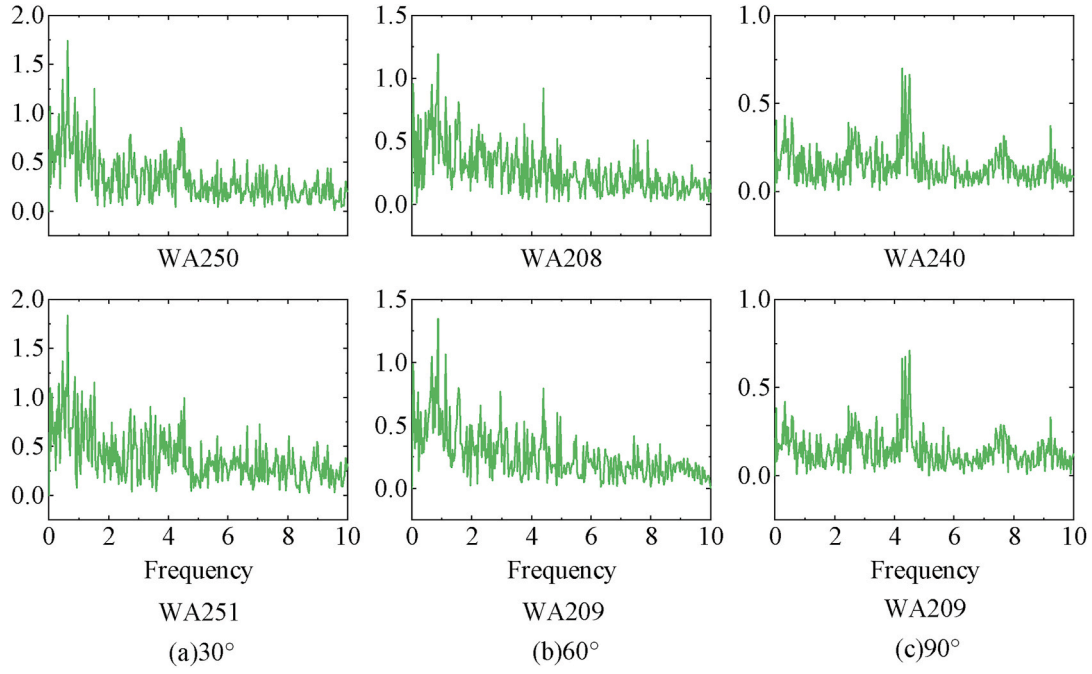


Fig. 14. Wind pressure frequency spectra: (a) 30°, (b) 60°, (c) 90°.

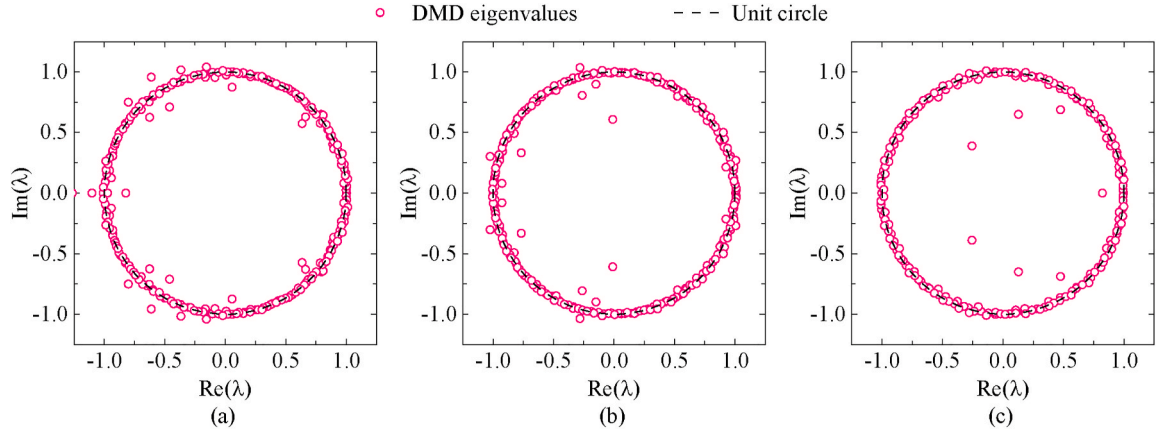


Fig. 15. Distribution of eigenvalues: (a) 30°, (b) 60°, (c) 90°.

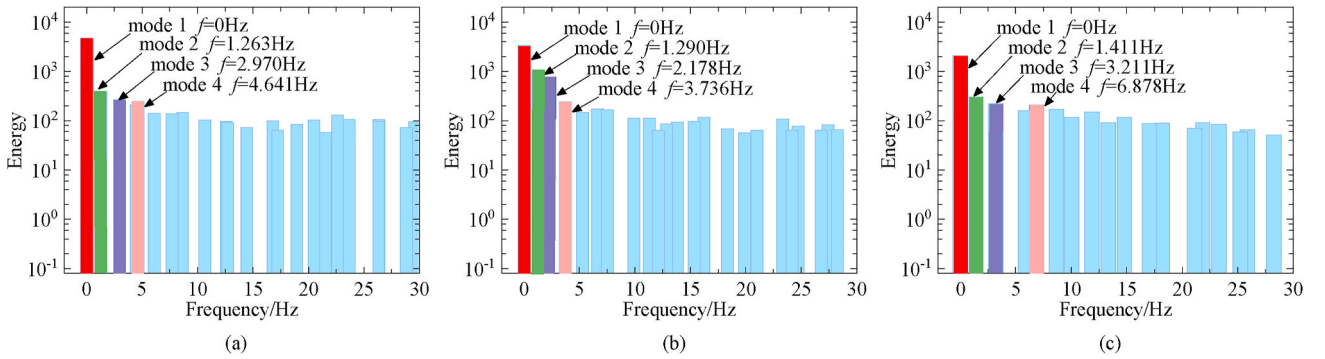


Fig. 16. Energy distribution: (a) 30°, (b) 60°, (c) 90°.

original data were sampled at intervals. Specifically, one data point was selected every nine points from the original data. Subsequently, the first five dominant modes were selected to analyze the distribution characteristics of the wind pressure coefficient at $\theta = 0^\circ$.

Fig. 6(a) exhibits the distribution of DMD eigenvalues at $\theta = 0^\circ$. The black dashed line represents the unit circle, while the pink circles denote the DMD eigenvalues. The real part of the eigenvalues is represented by the horizontal axis, corresponding to the growth rate (or decay rate) of

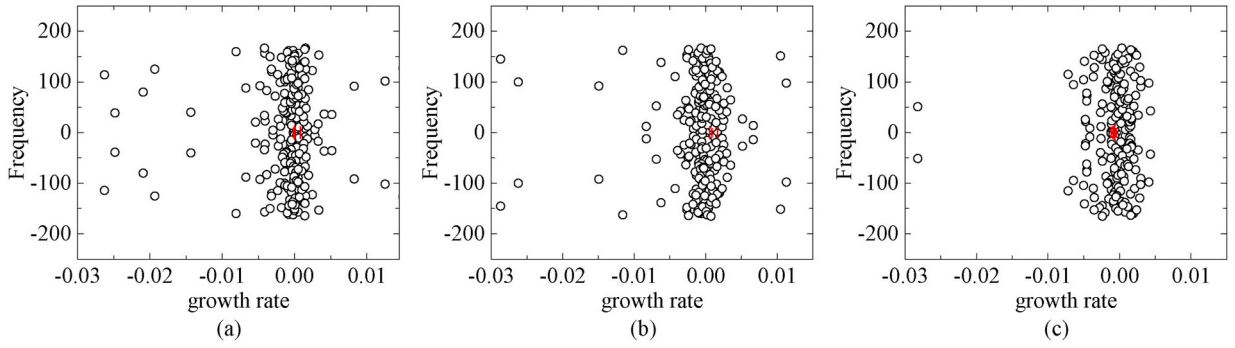


Fig. 17. The distribution of growth rate of DMD modes: (a) 30°, (b) 60°, (c) 90°.

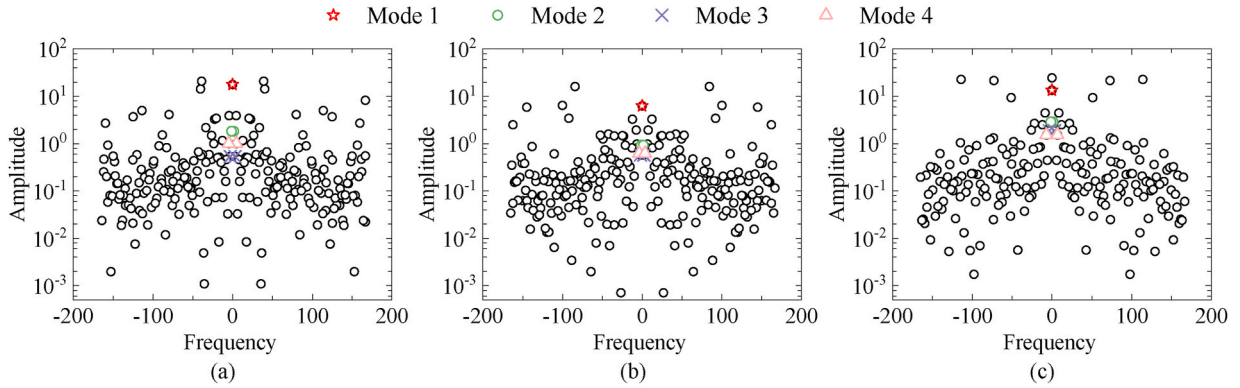


Fig. 18. The amplitude distribution of DMD modes: (a) 30°, (b) 60°, (c) 90°.

the modes. A positive value on the horizontal axis indicates that the corresponding mode is unstable, with its amplitude increasing over time, whereas a negative value indicates stability, with the amplitude decreasing over time. The imaginary part of the eigenvalues is represented by the vertical axis, corresponding to the frequency of the modes. The presence of the unit circle is used to represent the stability of the eigenvalues. If the eigenvalues within the unit circle depict stability, while those outside the unit circle indicate instability. From Fig. 6(a), it can be observed that the majority of eigenvalues are distributed on the unit circle. This suggests that these modes are stable.

The growth rate distribution of DMD modes at $\theta = 0^\circ$ is depicted in Fig. 6(b). The relationship between growth rate and frequency represents the temporal growth or decay of modes. When the growth rate approaches zero, it indicates that the mode undergoes minimal growth or decay over time, exhibiting relatively stable characteristics. The growth rate is an important parameter for selecting dominant modes, with dominant modes having growth rates close to zero. From Fig. 6(b), we can observe that most of the modes concentrate around growth rates close to zero, indicating the stability of the cylindrical coal shed pressure field.

Fig. 7 illustrates the first 5 dynamic modes of DMD at $\theta = 0^\circ$. Mode 1 is typically the mean mode, representing the background flow, corresponding to the mean of the flow field within the sampling section. Higher-order modes can be seen as perturbed flow fields, describing small-scale variations or disturbances relative to the mean mode. Additionally, higher-order modes exhibit relatively smaller flow structures.

Comparing Fig. 7 with Fig. 4(a) at $\theta = 0^\circ$, it is evident that the characteristic distribution of mode 1 obtained from the DMD decomposition is similar to the distribution of the mean wind pressure coefficient on the cylindrical surface. Mode 1 basically captures the overall trend of the mean wind pressure coefficient distribution across the cylindrical surface. However, the characteristic distribution of the other

modes differs greatly from that of mode 1, which may be physically attributed to the formation of flow separation vortices in the incoming airflow. As the modal order increases, the similarity between the feature distribution and the overall distribution of wind pressure coefficients decreases, with higher-order modes primarily reflecting local wind pressure distributions.

The reduced-order modes of DMD enable the reconstruction of the pressure field. The error between the original pressure field and the reconstructed pressure field is related to the number of modes, so it is evaluated by the loss function. The loss function is expressed as follows:

$$\text{loss} = \frac{\|\mathbf{CP}_1 - \Phi \mathbf{D}_\alpha \mathbf{V}_{and}\|_F}{\|\mathbf{CP}_1\|_F} \cdot 100\% \quad (17)$$

where Φ is the DMD eigenvector matrix $\Phi = [\Phi_1, \Phi_2, \dots, \Phi_r]$, $\mathbf{D}_\alpha$ represents the diagonal matrix of the amplitude $[\alpha_1, \alpha_2, \dots, \alpha_r]$, and $\mathbf{V}_{and}$ is the Vandermonde matrix.

Fig. 8 presents the relationship between losses and the number of modes at $\theta = 0^\circ$. With an increase in the number of modes, there is a tendency for the loss to decrease. Mode 1 already captures most of the information regarding the pressure field, and the first 5 modes recover approximately 85% of the information. After the number of modes reaches 100, within a small range, increasing the number of modes does not significantly improve the accuracy of reconstructing the pressure field. In this paper, the defined loss represents the relative error between the original pressure field and the reconstructed field for each time series. When a sufficient number of modes are included, the errors arising from the inconsistency between the loss and ordering methods can be mitigated.

To further assess the effectiveness of DMD in extracting the wind pressure distribution characteristics of cylindrical coal sheds, pressure field reconstruction is conducted using the first 5 DMD modes that were obtained. Fig. 9 illustrates the original and reconstruction pressure fields at $\theta = 0^\circ$. The left side represents the original pressure field, the middle

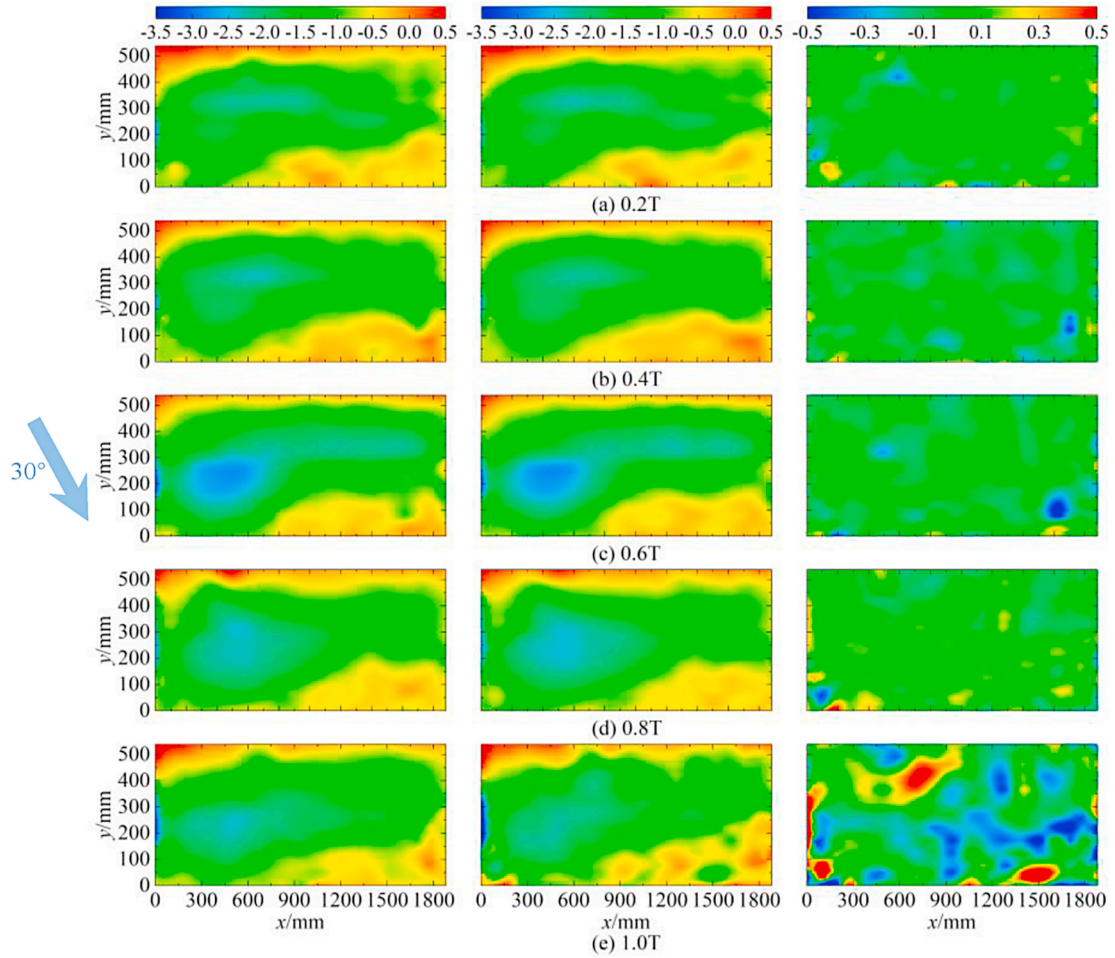


Fig. 19. The original and reconstructed pressure fields corresponding to the first 15 modes at $\theta = 30^\circ$ at five instants (Left: original pressure field; middle: reconstructed pressure field; right: absolute error of the reconstruction).

side depicts the reconstructed pressure field, and the right side shows the absolute error between the original and reconstructed pressure fields. The first 5 modes can preserve the primary dynamic characteristics of the data. Considering both efficiency and accuracy in reconstruction, we opt for utilizing the first 5 modes to effectively reconstruct the pressure field to capture the main structures. At $\theta = 0^\circ$, the leading edge AB of the cylindrical structure experiences an impact with the incoming flow, resulting in positive pressure with a maximum wind pressure coefficient of approximately 0.3. Subsequently, the airflow swiftly ascends along the curved surface, leading to separation near the roof, which predominantly falls within the negative pressure zone, enduring wind suction. In the wake region, the airflow bypasses the top of the structure and reattaches downstream, reducing negative wind pressure. However, significant errors during the reconstruction process, particularly at the instant of 0.8T and 1.0T, are clearly observable from the error contour plots in Fig. 9. To enhance reconstruction accuracy, it is contemplated to employ higher-order modes and reduce reconstruction errors, thereby further improving the reconstruction effectiveness.

Fig. 10 illustrates the first 20 modes of the DMD at $\theta = 0^\circ$, and we can see that the higher-order modes reflect the local wind pressure distribution. Figs. 11 and 12 illustrate the comparison between reconstructed pressure fields using different orders of DMD and the original pressure fields at $\theta = 0^\circ$. The left side depicts the original pressure fields, the middle side shows the reconstructed pressure fields and the right side represents the absolute error between them. By examining the error contour plots in Figs. 11 and 12, it is evident that the errors have been significantly reduced. This indicates that employing DMD for

reconstruction can effectively enhance the accuracy, rendering the reconstructed results closer to the original pressure fields. It is noteworthy that in Fig. 11, the reconstructed results using 15 orders of DMD are already quite close to the original pressure fields, suggesting that this order of reconstruction is adequate to meet the requirements. The utilization of high-order DMD modes for reconstruction has successfully enhanced the accuracy of pressure field reconstruction. These results provide a reliable foundation for further analysis and application.

To further assess the effect between the number of modes and the reconstruction effect, the accuracy of the reconstructed pressure field is quantified using the relative error of the reconstruction at different moments.

$$Error(rec_i) = \frac{\|v_i^{ori} - v_i^{rec}\|_2}{\|v_i^{ori}\|_2} \cdot 100\% \quad (18)$$

where $\|v_i^{ori} - v_i^{rec}\|_2$ is the L_2 -norm of the deviation between the original and reconstructed fields, $\|v_i^{ori}\|_2$ is the L_2 -norm of the original field, and $\|v_i^{rec}\|_2$ is the L_2 -norm of the reconstructed field.

The reconstruction relative error of the modal reconstruction with different orders is presented in Fig. 13. It can be observed that higher-order DMD reconstructions can effectively improve the reconstruction accuracy. The accuracy of the reconstruction using 15 modes is significantly better than that of the 5 modes of reconstruction. Additionally, increasing the number of modes to 16, 18, or 20 did not significantly improve the reconstruction accuracy. Further increasing the number of modes may not significantly improve the reconstruction accuracy.

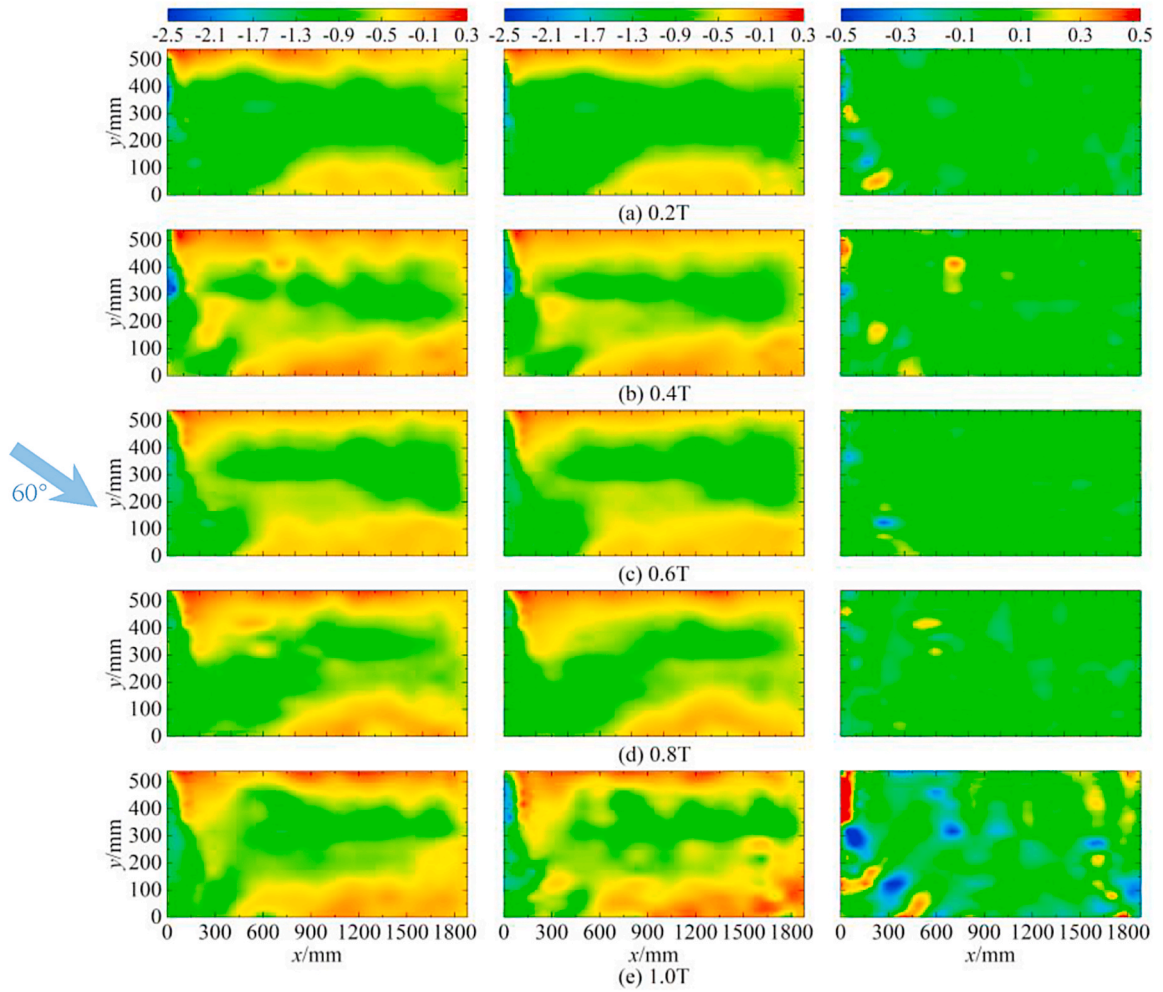


Fig. 20. The original and reconstructed pressure fields corresponding to the first 15 modes at $\theta = 60^\circ$ at five instants (Left: original pressure field; middle: reconstructed pressure field; right: absolute error of the reconstruction).

Therefore, the reconstruction using 15 modes is sufficient to meet the required accuracy.

3.3. DMD analysis and reconstruction under other wind directions

The wind pressure frequency spectra at $\theta = 30^\circ$, 60° , and 90° are depicted in Fig. 14, corresponding dominant frequencies are 0.65 Hz, 0.87 Hz and 4.3 Hz, respectively, with cycles of 1.54s, 1.15s and 0.23s, respectively. To improve the reconstruction and prediction accuracy of DMD, five complete cycles of data at each wind direction were selected for analysis, corresponding to a total time of 7.7s, 5.75s and 1.15s, respectively. In each cycle, the data of 50 time points are included, so the pressure field data of $T = 250$ time points are selected in total. During data processing, to ensure that the time interval of the pressure field data matches the time interval of the snapshot, the original data is processed by sampling at intervals. Therefore, one data point is taken from the original data at intervals of 11, 8 and 2 data points, respectively.

Based on the performance of three reconstruction methods at $\theta = 0^\circ$, using the first 15 modes of DMD, the wind pressure fields at $\theta = 30^\circ$, 60° , and 90° were decomposed and reconstructed. The distribution of DMD eigenvalues and their energy distribution at $\theta = 30^\circ$, 60° , and 90° are illustrated in Figs. 15 and 16, respectively. Fig. 15 shows that most of the eigenvalues are distributed on the black dotted line, indicating the stability of DMD modes at different wind directions. Fig. 16 depicts a dense frequency distribution along the cylindrical coal shed, with the cumulative energy of the first four modes accounting for 32.22%, 36.17%, and

25.75%, respectively. Mode 1 respectively represents 26.34%, 19.84%, and 18.42% of the total pressure field energy.

Fig. 17 displays the distribution of growth rate of DMD modes at $\theta = 30^\circ$, 60° , and 90° . The most of modes are concentrated around zero growth rate. Fig. 18 illustrates the amplitude distribution of DMD modes at $\theta = 30^\circ$, 60° , and 90° . In this paper, the individual modes are sorted based on their amplitudes. The colored dots in the figure indicate the first 4 orders of dynamic modes, and each black dot represents the corresponding mode amplitude of the pressure field. The first mode extracted by DMD is mode 1, while the remaining modes appear in conjugate complex pairs, thus each pair can be considered as the same mode. It can be observed from Fig. 18(a) and (b) that the ordering of modes 3 and 4 has changed. Although the sorting method used in this study orders the DMD modes based on their amplitudes, there may be certain limitations associated with this approach. Therefore, in the next step, we plan to employ alternative sorting methods to enhance the accuracy of mode ordering, enabling a better understanding and interpretation of the relationships between different modes.

Fig. 19 compares the original and reconstructed pressure fields at $\theta = 30^\circ$. The left side depicts the original pressure field, the middle side displays the reconstructed pressure field, and the right side demonstrates the absolute error between them. At $\theta = 30^\circ$, flow separation occurs in the region of the cylindrical edge AC, leading to the formation of a significant adverse pressure gradient zone. Consequently, the negative wind pressure coefficient reaches its maximum value in this region, making it a wind-sensitive area. Therefore, when conducting

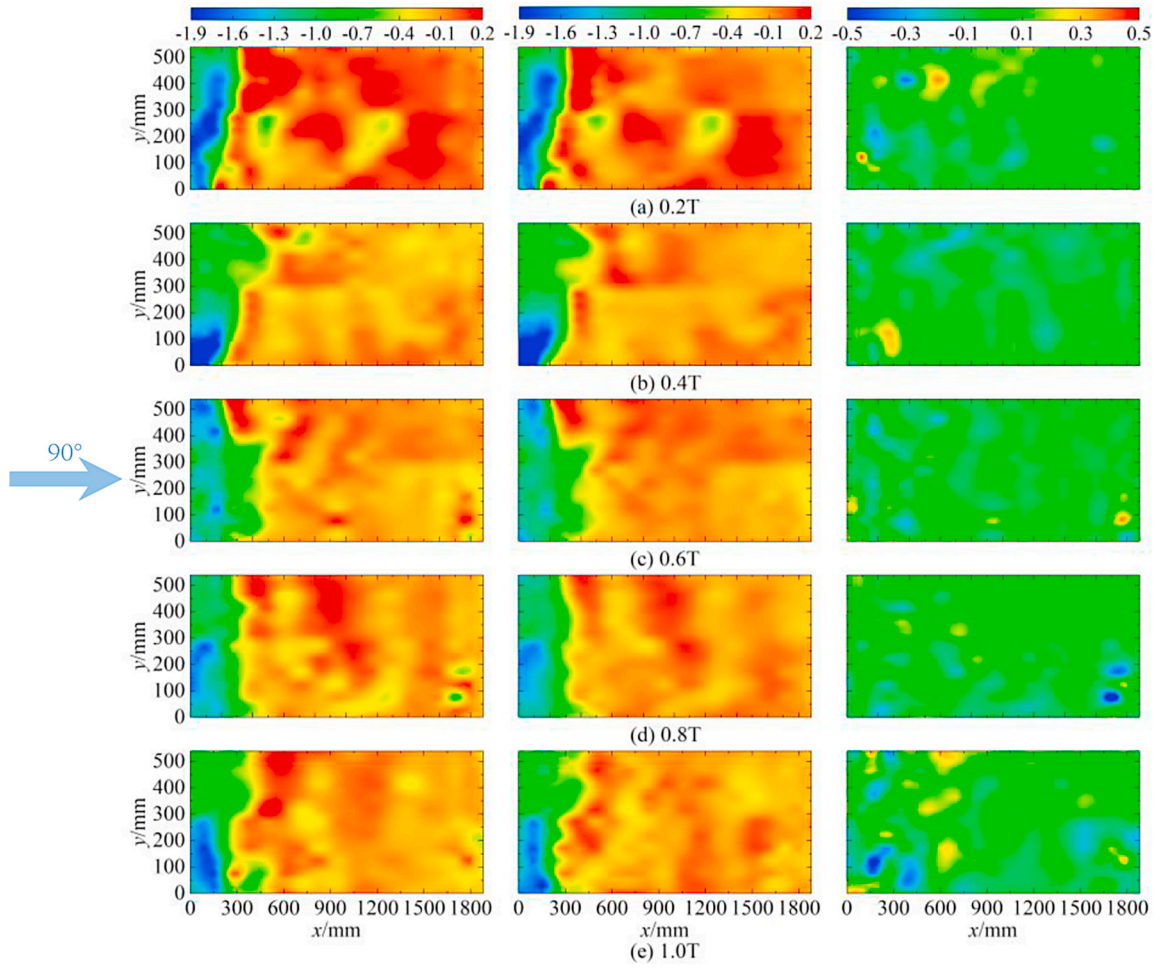


Fig. 21. The original and reconstructed pressure fields corresponding to the first 15 modes at $\theta = 90^\circ$ at five instants (Left: original pressure field; middle: reconstructed pressure field; right: absolute error of the reconstruction).

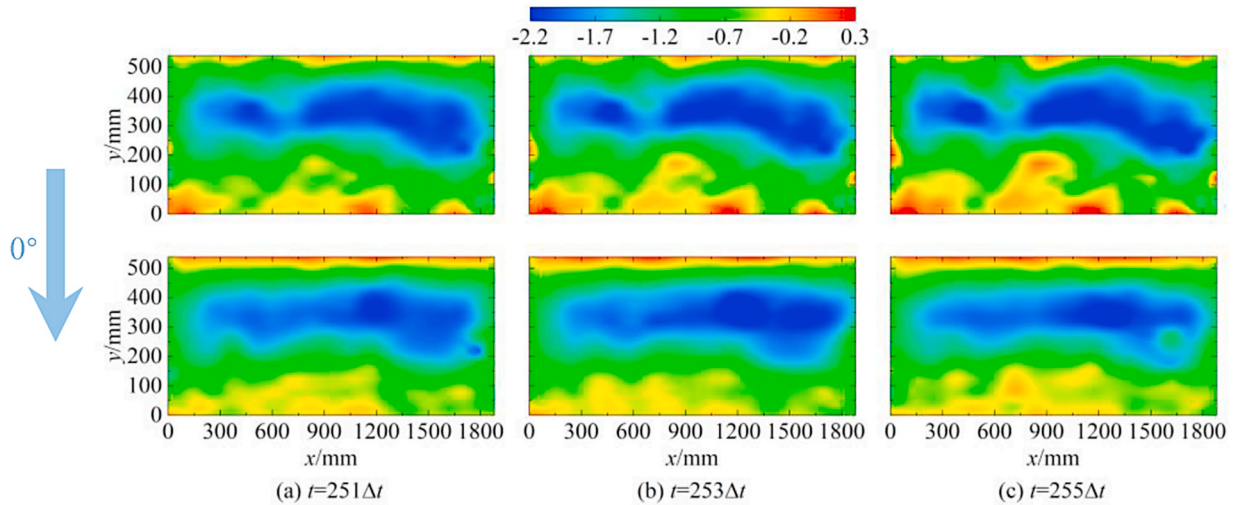


Fig. 22. Prediction of wind pressure distribution at $\theta = 0^\circ$ (Top: prediction pressure field; bottom: original pressure field).

wind resistance calculations for large-span structures, it is advisable to implement protective measures at the edges of the roof. In this case, both positive and negative values of the cylindrical wind pressure coefficient coexist, with the lower-right region, far from the incoming flow,

predominantly exhibiting positive pressure. It is worth noting that the reconstruction error gradually increases over time, particularly at the instant of 1.0T. This could be attributed to the sorting of DMD modes in this study based on their modal amplitudes, which inherently limits this

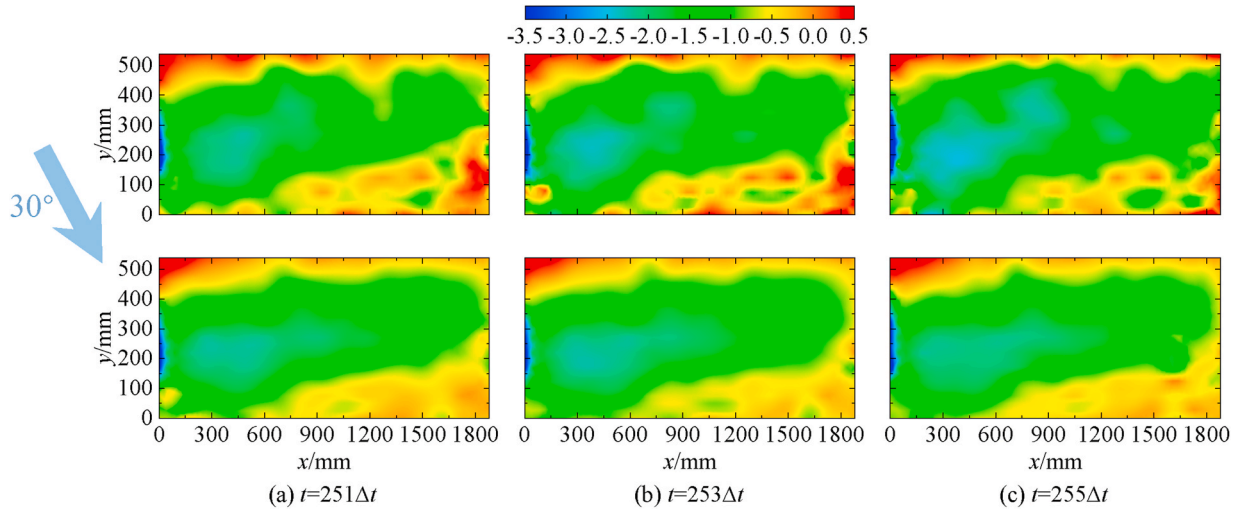


Fig. 23. Prediction of wind pressure distribution at $\theta = 30^\circ$ (Top: prediction pressure field; bottom: original pressure field).

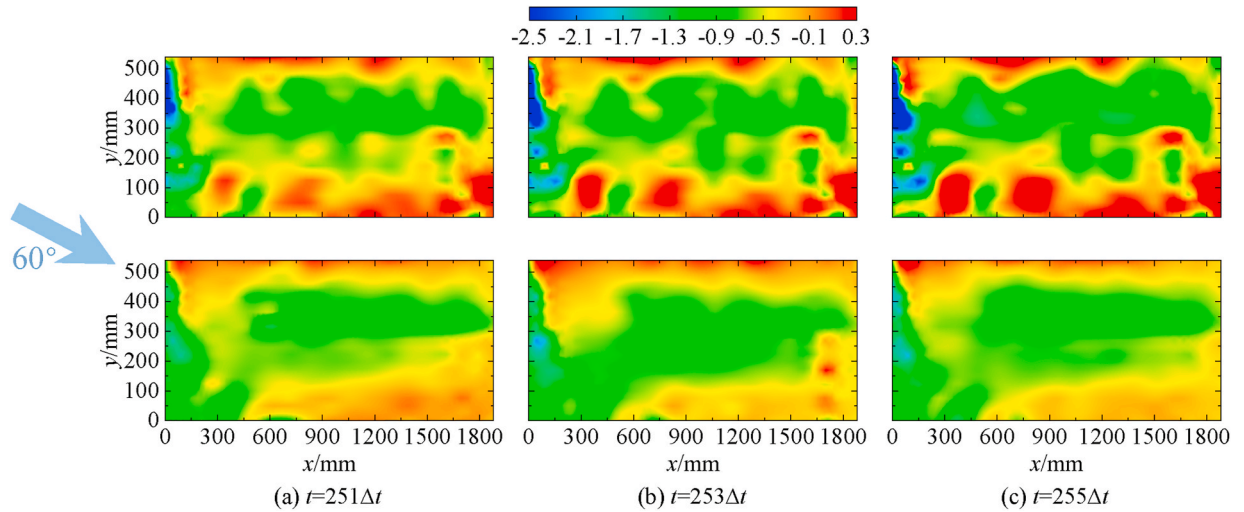


Fig. 24. Prediction of wind pressure distribution at $\theta = 60^\circ$ (Top: prediction pressure field; bottom: original pressure field).

sorting method. The dynamic characteristics of the airflow pose a greater challenge for reconstruction, resulting in larger reconstruction errors. Therefore, during pressure field reconstruction, special attention needs to be paid to the unfavorable wind direction of 30° , and appropriate measures should be taken to reduce reconstruction errors.

Fig. 20 depicts a comparison between the original pressure field and the reconstructed pressure field at $\theta = 60^\circ$. The left side depicts the original pressure field, the middle side displays the reconstructed pressure field, and the right side demonstrates the absolute error between them. At $\theta = 60^\circ$, the maximum value of the negative wind pressure coefficient appears on the edge of the AC of the cylinder, accompanied by significant flow separation. A large negative pressure gradient area is formed, constituting a wind-sensitive region. Positive pressure is predominantly observed near the edges of AB and CD. The regions parallel to line AB on both sides of the cylindrical structure exhibit predominantly positive pressure. Similar to the case at $\theta = 30^\circ$, the reconstruction error gradually increases over time, especially at the instant of $1.0T$. Therefore, special attention should be given to the unfavorable wind directions of both 30° and 60° , and corresponding measures should be taken to reduce the reconstruction error, enhance the reconstruction accuracy, and improve the overall reconstruction precision.

Fig. 21 illustrates a comparison between the original pressure field and the reconstructed pressure field at $\theta = 90^\circ$. The left side depicts the original pressure field, the middle side displays the reconstructed pressure field, and the right side demonstrates the absolute error between them. When airflow passes over the roof along the EF direction, due to the enclosed nature of the model, the airflow collides with the gable wall and subsequently bypasses the edge AC region, flowing towards the rear. This results in a relatively high absolute value of negative pressure in the lower-left region of the roof. Subsequently, the airflow exhibits reattachment behavior in the middle and rear parts of the roof surface, causing a relatively small variation in the mean wind pressure coefficient in these areas. In the lower-left region, close to the incoming flow, the reconstruction error is relatively significant. Therefore, at $\theta = 90^\circ$, special attention should be given to areas where the airflow directly collides, and optimization designs or appropriate protective measures should be implemented to reduce the potential risk of structural damage.

3.4. Short-term prediction results

DMD captures the dynamic behavior of a system from time-series data and decomposes it into a series of modes, each of which

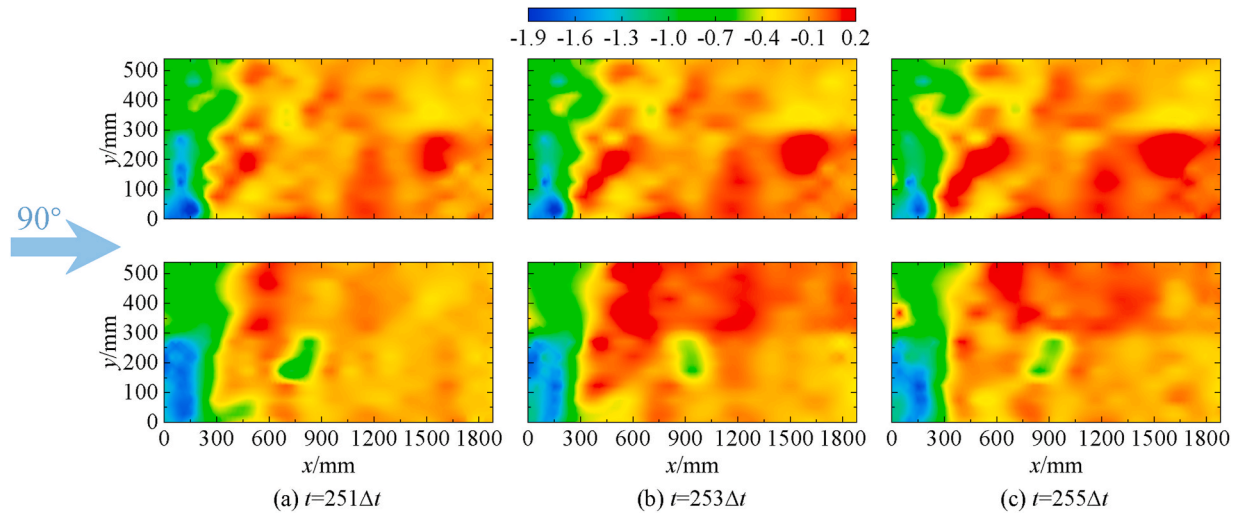


Fig. 25. Prediction of wind pressure distribution at $\theta = 90^\circ$ (Top: prediction pressure field; bottom: original pressure field).

corresponds to a specific vibration pattern or feature within the system, thus describing the fundamental characteristics of the system's evolution over time. Additionally, DMD provides dynamic features of the system, such as modal growth rates and frequencies. These dynamic characteristics can aid in predicting future system behavior, making DMD suitable for short-term prediction (Chen et al., 2023). The reduced-order DMD algorithm was employed to predict the short-term wind pressure distribution of the cylindrical coal shed at $\theta = 0^\circ$, 30° , 60° , and 90° , with the results depicted in Figs. 22–25.

From the figures, it is evident that the DMD algorithm can effectively predict the pressure field over a short period at these wind directions, making it applicable for time-series prediction of wind pressure distribution in the coal shed. The predictive performance is relatively good for positions near the incoming flow, but less accurate for locations further away from the incoming flow, particularly exhibiting poor performance at $\theta = 60^\circ$. Moreover, as time progresses, the prediction errors accumulate, especially at positions away from the incoming flow. These prediction errors may stem from various factors, including data quality, prediction time span, and system complexity. Thus, when conducting predictions, it is crucial to comprehensively consider these factors and take further measures to enhance prediction accuracy. In the next step, considering the incorporation of additional physical information could improve the model's predictive capabilities. By combining physical information with data-driven methods, the accuracy and reliability of the prediction model can be enhanced, better addressing the challenges of complex systems. Moreover, the DMD algorithm used in this study has some limitations. Specifically, the method employed ranks DMD modes based on mode amplitudes, a technique that may not be ideal in certain scenarios. Kou and Zhang (2017) proposed a new criterion for selecting dominant modes using DMD, which considers the evolution of each mode across the entire sampling space. This approach ranks modes according to their contribution to all sampling points, thereby providing a more accurate basis for mode selection. Additionally, the DMD algorithm in this study does not effectively filter out noise introduced by instrument errors or data processing. Consequently, unnecessary interference signals may remain in the raw data, potentially impacting the accuracy and reliability of the results. To address this, future research will focus on exploring improved DMD algorithms that enhance robustness against noise. For example, Dawson et al. (2016) proposed a DMD algorithm that corrects for the effect of sensor noise, while Le Clainche and Vega (2017) proposed TDMD and high-order DMD (HODMD), which can effectively remove noise bias and thus improve data accuracy and reliability.

4. Conclusions

This study initially examines the wind pressure distribution characteristics of cylindrical coal sheds at varying wind directions. Subsequently, DMD is employed to decompose the wind pressure field of undisturbed coal sheds at different wind directions into modes, and a reduced-order model is established utilizing the principal modes obtained from the decomposition. An analysis of the reconstruction accuracy and errors of the wind pressure field is conducted. Finally, DMD is utilized to predict the short-term wind pressure distribution of coal sheds at different wind directions. The main conclusions are summarized as follows:

- (1) The eigenvalues obtained from the DMD decomposition mostly lie within or on the unit circle, representing that the pressure field is stable. And dynamic behavior of the system can be effectively described by a limited number of modes. Each mode obtained by DMD corresponds to different frequency, amplitude, and temporal dynamic features. Among them, mode 1 is the largest in the proportion of the total energy and has a lower frequency. Mode 1 is usually similar to the mean pressure field and reflects the dominant pressure distribution characteristics in the pressure field. Higher-order modes correspond to transient pressure fluctuations at higher frequencies and are able to reveal more complex dynamic features.
- (2) The reconstructed results demonstrate that the reduced-order model based on DMD can effectively restore the pressure field. And as the number of modes increases, the reconstruction error gradually decreases. The reconstruction error mostly concentrates on the separation point and the wake region and exhibits strong nonlinearity.
- (3) At various wind directions, there are reconstruction errors in the pressure field, particularly noticeable at $\theta = 30^\circ$ and 60° where the errors are more significant. This observation may be attributed to the presence of separation zones and regions with negative pressure gradients.
- (4) DMD can be utilized for short-term predictions, with better predictive performance observed for positions closer to the inflow. Our next step involves integrating physical insights to enhance the predictive accuracy and effectiveness of DMD.

Large-span roof structures have dense frequency distributions, making DMD an effective method for analyzing the wind pressure

distribution on cylindrical coal sheds. Moreover, reduced-order models based on DMD are useful for predicting the surface wind pressure on coal sheds. In future work, we aim to delve deeper into the DMD algorithm and explore reduced-order reconstruction of wind pressure distribution under varying working conditions. By optimizing the sorting methods and involving higher-order modes, we aim to reduce the reconstruction error, improve the reconstruction accuracy, and capture flow details more effectively. We plan to further analyze wind pressure distribution characteristics, focusing on separation zones and negative pressure gradient areas, using the involvement of higher-order modes to interpret these features. Simultaneously, we will consider combining physical information and data-driven methods to improve the accuracy and reliability of prediction models, thereby better addressing the challenges posed by complex systems.

CRediT authorship contribution statement

Qingquan Liu: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Jing Huo:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shijie Liu:** Formal analysis, Data curation. **Zejun Qin:** Investigation, Formal analysis. **Haohan Li:** Investigation, Formal analysis. **Zhen Zhang:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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